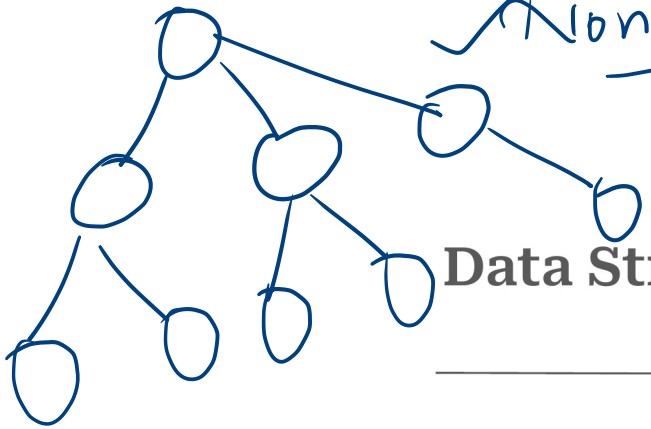
✓ Linear Data Structure ⇒

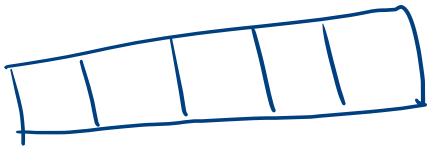
✓ Non " " " ⇒ Trees & Graphs



Data Structures and Algorithms

Trees

Mentor: **Adarsh Gupta**
(Senior Software Engineer)

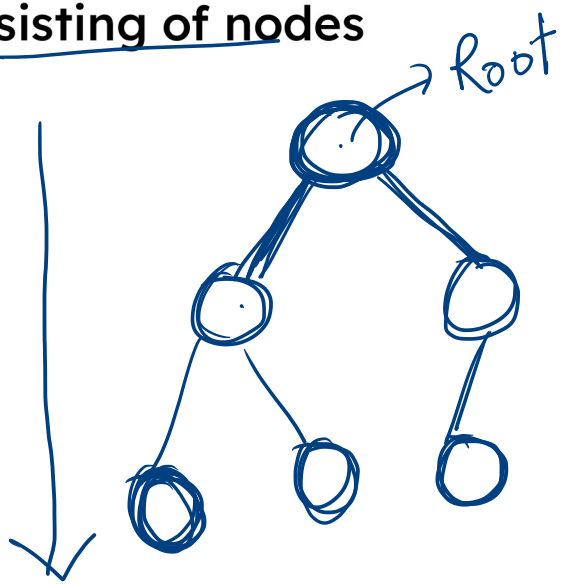




Definition:

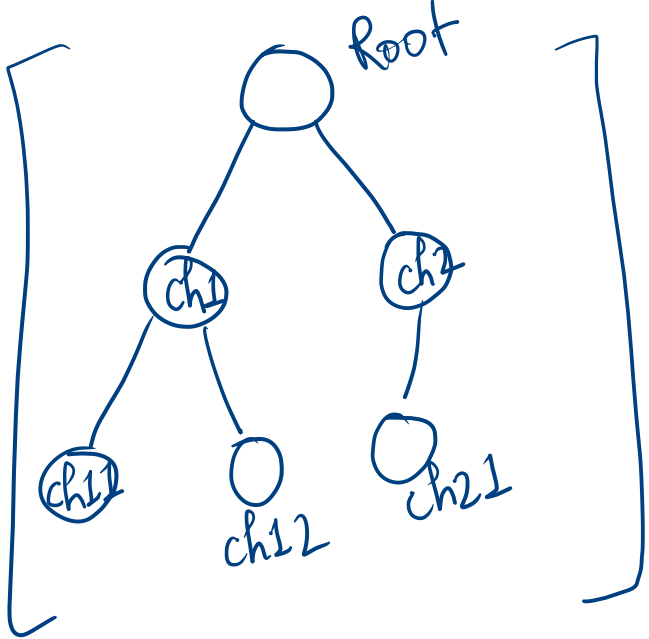
A Tree is a non-linear, hierarchical data structure consisting of nodes connected by edges.

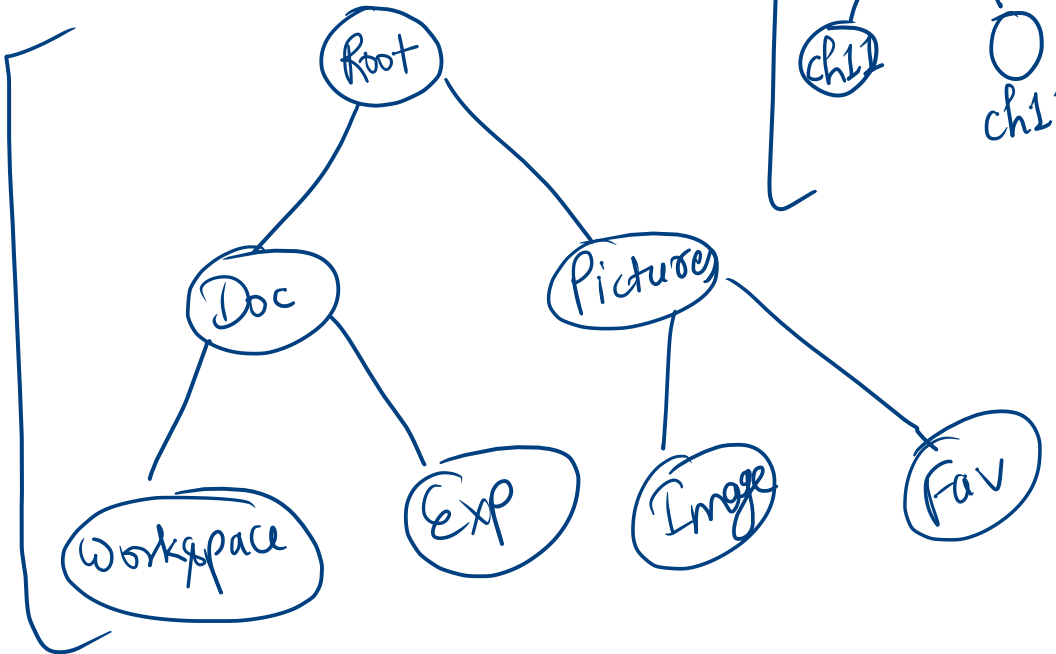
- A tree starts with a root node.
- Each node can have zero or more children.
- The top-down representation forms a hierarchy.



Real Life Examples:

- **Family Tree**
- **Folder Structure**

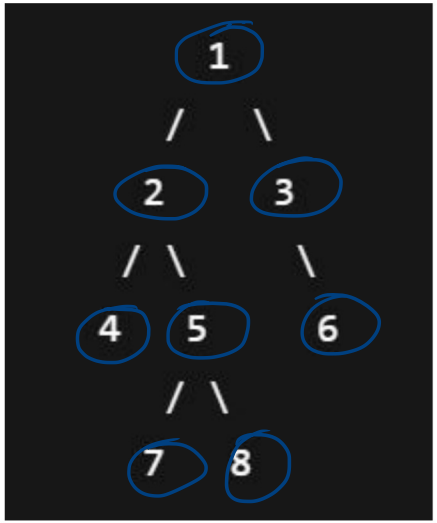




Basic Terminologies:

Nodes : A node is the basic unit of a tree that contains data and links (references) to its child nodes.

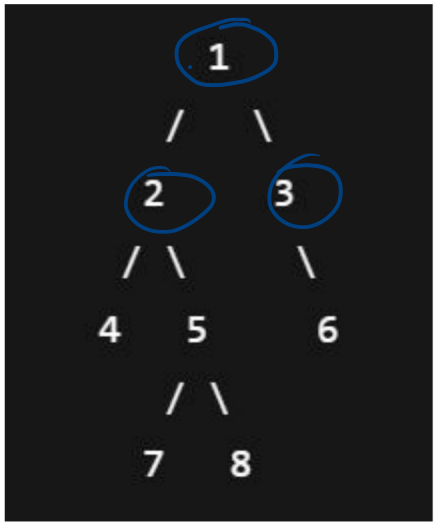
Nodes: 1, 2, 3, 4, 5, 6, 7, 8



Basic Terminologies:

Root Node : The topmost node in a tree which does not have any parent. Every tree has exactly one root node.

Root Node : 1

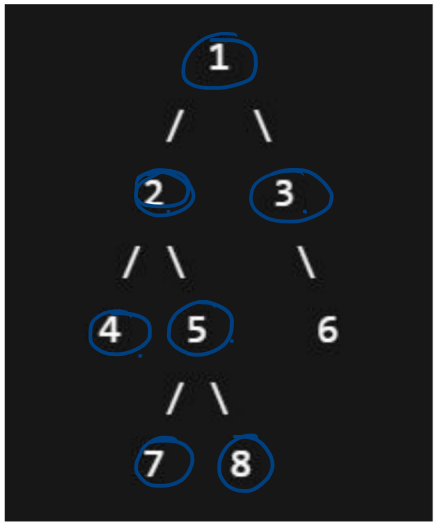


Tree

Basic Terminologies:

Parent Node : A node that has one or more child nodes connected to it.

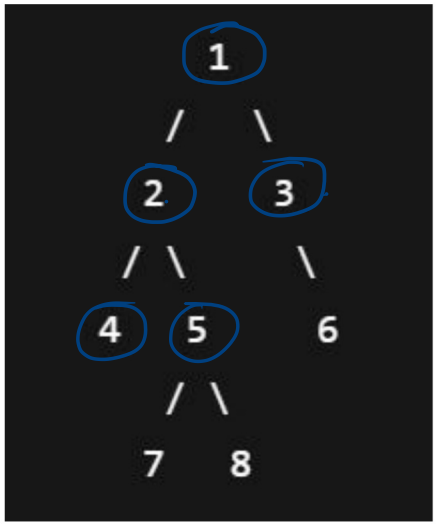
Parent of 2 & 3 : 1
Parent of 4 & 5 : 2
Parent of 7 & 8 : 5



Basic Terminologies:

Child Node : A node that descends from another node (its parent).

Children of 1 : 2 & 3
Children of 2 : 4 & 5

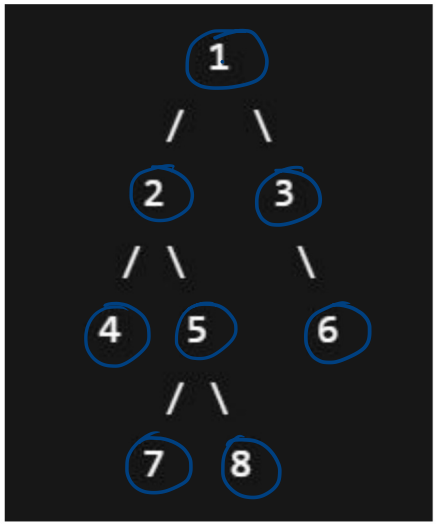


Basic Terminologies:

→ [External Node]

Leaf Node : A node that has no children.

Leaf Nodes are: 4, 6, 7, 8

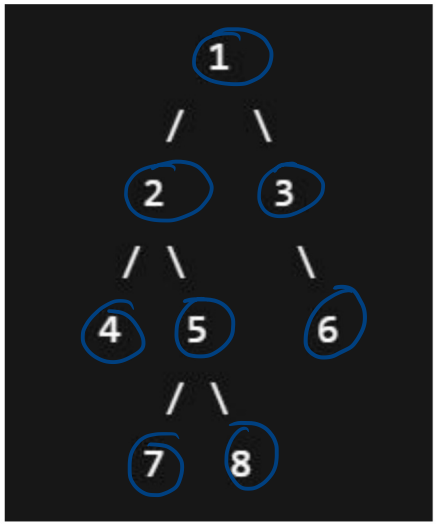


Tree

Basic Terminologies:

Internal Node (Non-leaf Node) : A node that has at least one child.

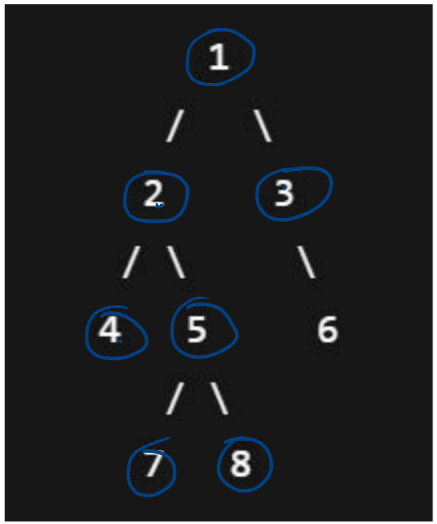
Non-Leaf Nodes are : 1, 2, 3, 5



Basic Terminologies:

Siblings : Nodes having the same parent are called siblings.

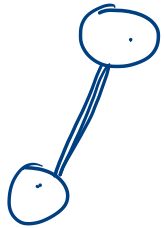
- 2 and 3 are siblings (same parent 1).
- 4 and 5 are siblings (same parent 2).
- 7 and 8 are siblings (same parent 5).



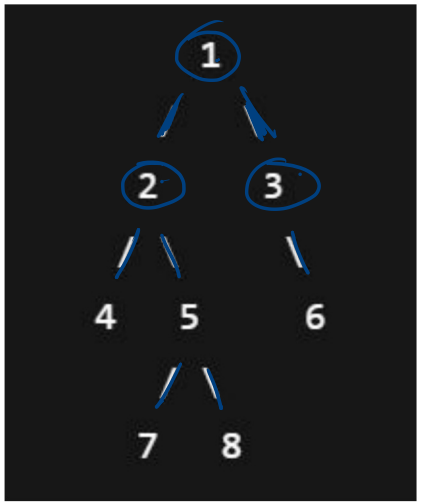
Tree

Basic Terminologies:

Edge: A connection between parent and child.
Every edge connects two nodes.



Edges are: (1-2), (1-3),
(2-4), (2-5), (3-6),
(5-7), (5-8)



Tree

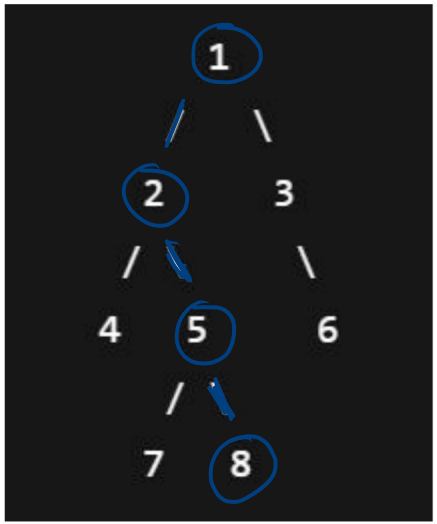
Total Edges = No. of Nodes - 1
= 8 - 1
= 7

Basic Terminologies:

Path: A sequence of nodes connected by edges from one node to another.

Path from 1 to 8 is →

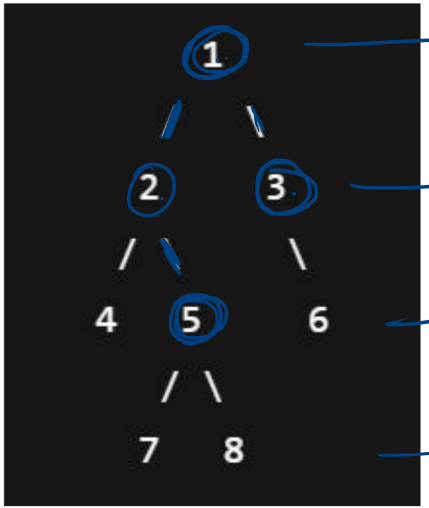
1 → 2 → 5 → 8



Basic Terminologies:

Level of a Node: The number of edges in the path from the root to that node.
The root node is at level 0.

Level of 8 ⇒ 3



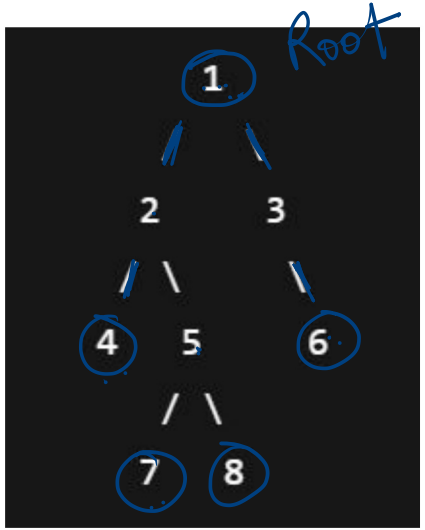
Basic Terminologies:

Height of a Tree: The number of edges in the longest path from root to leaf.

Longest path = 1 → 2 → 5 → 7
Height = 3

root leaf
1 → 2 → 4
1 → 3 → 6
1 → 2 → 5 → 7
1 → 2 → 5 → 8.





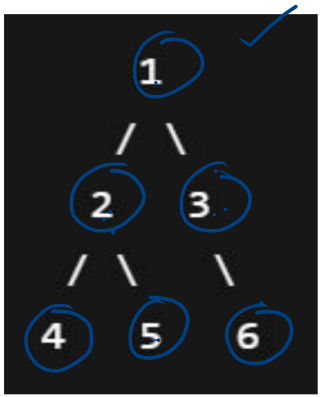
Height of this tree = 3

Types of Tree Data Structure:

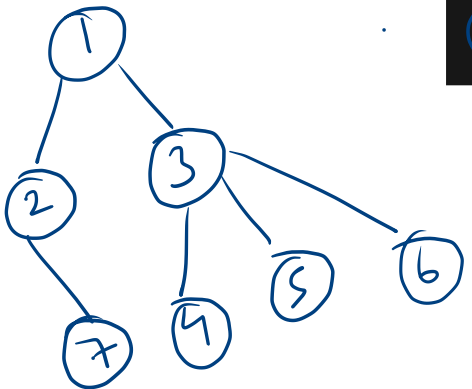
Binary Tree: A Binary Tree is a type of tree data structure in which each node can have at most two children — referred to as the left child and right child.

0, 1, 2

Not more than 2



Binary Tree



Is this B.T.?

No

Binary Tree

How to create Node in Java

```
class Node {
    int data;
    Node left, right;

    Node(int value) {
        data = value;
        left = right = null;
    }
}
```

Node left ;
Node right ;

Constructor → To initialize

